

OpenTURNS release highlights

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User Day #18, June 12th 2026, EDF Lab

AIRBUS



New features since last year in releases:

- v1.26: fall 2025
- v1.27: spring 2026

- 1 Finalized new GPR API migration
- 2 New experimental classes
- 3 Misc

Gaussian process regression API

Better hyperparameter optimization (1.26)

- Normalization and improved scale bounds initialization
- Faster and more robust convergence

Heteroskedastic noise (1.27)

- Input-dependent noise in GPR
- New accessors: `getCholeskyFactor`, `getStandardizedOutput`

Cross-validation (1.27 / experimental)

- `GaussianProcessRegressionCrossValidation`
- Automatic hyperparameter selection
- Reworked GPR & chaos documentation

HiGHS linear solver

- Open-source high-performance LP solver



LinearProblem

- Framework for solving linear problems

FiniteOrthogonalFunctionFactory

- Build finite orthogonal function bases

- Improved functions API with graphs
- Improved documentation of Directional sampling classes
- Consistency of examples
- SVG graphics
- New example: pitfalls in polynomial chaos expansion
- Improved Pearson and Spearman theory files

Other improvements

- Smolyak nested tensorization
- Advanced validation of Distribution classes
- Allow MultiStart optimization in FORM algorithms
- Deepcopy support for all objects
- FunctionalChaosResult.getMarginal
- KDTree radius search
- Mesh convex flag
- TruncatedDistribution handles KernelMixture
- Centered data in Linear/QuadraticLeastSquares
- QuantileConfidence interval coverage
- KFoldSplitter non-interlaced folds
- Pagmo incremental evolution callback
- SpecFunc: BesselInu/LogBesselInu/Clip



Classes that left the experimental module:

1.26:

- LatentVariableModel
- UniformOrderStatistics
- RankSobolSensitivityAlgo
- ExperimentIntegration
- CubalIntegration
- SamplePartition
- GeneralizedParetoValidation
- FunctionalChaosValidation
- PointToFieldFunctionalChaosAlgorithm

1.27:

- PointConditionalDistribution
- GaussianProcessFitter
- GaussianProcessRegression
- GaussianProcessConditionalCovariance
- PosteriorDistribution

Deprecated (1.27):

- KrigingAlgorithm
- KrigingRandomVector
- GeneralLinearModelAlgorithm

Python channels

- Pip / Conda
- Versions: 3.9+ (abi3)
- OS: Windows, Linux, MacOS
- Architectures: x86_64, arm64 (Linux+MacOS)
- Nightly channel: pypi.anaconda.org/openturns-wheels-nightly

Supported Linux distributions

- Ubuntu 22/24/26
- Debian 11/12/13
- Fedora 43/44
- OpenSUSE 16.0
- Mageia 9
- ArchLinux

... and other various repositories

- FreeBSD
- NixOS
- Spack
- Guix
- Vcpkg

2026 plans

- Functional chaos expansion API improvements
- Classification with random forests
- LMG sensitivity indices (dependent variables for linear models)
- VonMises-Fisher distribution in high dimension
- LLM code review

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categories ▶

tags ▶

Categories

Latest

Top

+ New Category

✎ New Topic

Category

Topics

Latest

■ Development

29

Everything about hacking the library.

■ Announcements

29

Infos on new releases, events, etc

■ Python usage

89

Get help using openturns from its Python module

■ Methodology

45

Discussion on the uncertainty quantification methodology in studies

■ Installation

10



Sphinx gallery issue

■ Development

1

1d



OpenTURNS Users Day #18, Friday 13, June 2025

■ Announcements

0

Apr 16



Can an OpenTURNS PythonFunction transmit exceptions from the Python function?

■ Python usage

4

Mar 13



Is it possible to correlate two distributions in a BayesDistribution?

4

Feb 11

Thank you for your attention!
Any questions?

